

**Visvesvaraya Technological University,
Jnana Sangama, Belagavi – 590 018**



**KLS Vishwanathrao Deshpande Institute of Technology,
Haliyal – 581 329**

(Accredited by NAAC with “A” Grade)



Mini Project Report

Body Posture Monitor For Desk Workers

Submitted in partial fulfillment of the requirement for the award of degree of

Bachelor of Engineering

in

Electronics and Communication Engineering

Submitted by

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V – Semester / III – Year

Under the guidance of

Dr. Nagaraj Bhat

2025 – 26

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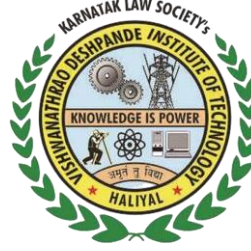
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CERTIFICATE

Certified that the Mini Project Work entitled “ **Body Posture Monitor for Desk Workes**” carried out by

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are Bonafide Students of KLS Vdit, Haliyal, in partial fulfillment for the award of Bachelor of Engineering in Electronics and Communication Engineering, of the Visvesvaraya Technological University, Belagavi during the Year 2025-2026. It is certified that all corrections / suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the Departmental Library. The Mini Project report has been approved as it satisfies the academic requirements in respect of Mini Project Work prescribed for the said Degree.

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CHAPTER 1:

INTRODUCTION

1.1 Introduction

In today's digital era, many people spend long hours sitting at desks, often in front of computers. Poor sitting or standing posture over prolonged periods can lead to back pain, spinal misalignment, and other health issues. Maintaining correct posture is essential for the overall health and productivity of desk workers, students, and professionals. With the advancement of sensor-based technology and affordable microcontrollers, it is now possible to develop compact devices that can monitor posture and alert users in real time.

1.2 Motivation

Observations show that most individuals working for extended hours at desks are unaware of their body posture. Continuous bending or slouching causes strain on the neck and spine. This motivated the development of a small, wearable, and cost-effective device that can detect improper posture and remind users to correct it, thereby improving health and comfort

1.3 Objectives

- 1.To design a wearable system for detecting incorrect sitting posture.
- 2.To Interface Flex sensor with ESP32 using MPU6050 for quaternion-based tracking.
- 3.To generate alerts using vibration motor when posture deviates beyond threshold.
- 4.To ensure portability by integrating the system into a light weight jacket.
- 5.To promote better ergonomics for desk workers and reduce health risks.

1.4 Scope of the Project

The proposed system is a portable, low-cost posture monitoring device using an ESP32 microcontroller and an MPU6050 sensor. It can be used by anyone who spends long hours sitting or standing in a fixed position. The project also provides a base for future enhancement, such as adding IoT features for mobile notifications, data logging, and posture-tracking analytics.

CHAPTER 2:

LITERATURE SURVEY

1. UWB-Posture Guard: A Privacy-Preserving RF Sensing System for Continuous Ergonomic Sitting Posture Monitoring

Authors: Haotang Li et al

Explanation:

- **Contactless Monitoring:** Uses ultra-wideband (UWB) RF signals to sense posture without cameras or wearables, ensuring strong privacy.
- **Temporal Classification:** Introduces a gradient-boosted decision-tree model (PoseGBDT) for analyzing continuous posture sequences.
- **High Accuracy:** Achieves ~99% accuracy across 19 sitting postures, robust against clothing, devices, and environmental changes.
- **Real-World Usability:** Designed for continuous, long-term ergonomic monitoring in offices while minimizing invasiveness.

2. Sitting Posture Recognition and Feedback: A Literature Review

Authors: Christian Krauter et al.

Explanation:

- **Comprehensive Survey:** Reviews 223 papers related to sitting posture detection and feedback mechanisms.
- **Common Technologies:** Finds that pressure sensors and visual feedback systems are the dominant sensing and feedback methods.
- **Design Trade-offs Identified:** Highlights challenges such as privacy, accuracy, cost, and user comfort across different approaches.
- **Gap in Long-Term Studies:** Notes lack of consistent evaluation methods and calls for long-term, real-world posture studies.

3. Sensor-Based Wearable Systems for Monitoring Human Motion and Posture: A Review

Authors: Xinxin Huang et al.

Explanation:

- **Wearable Focus:** Provides a wide review of wearable sensors (IMUs, accelerometers, gyros, flexible sensors) used for motion/posture monitoring.
- **Sensor Placement Analysis:** Discusses optimal placement (waist, trunk, limbs) and how placement affects accuracy and comfort.
- **System Design Discussion:** Covers complete pipeline: sensing, transmission, processing, and real-time monitoring challenges.
- **Future Challenges:** Identifies accuracy, durability of flexible sensors, and data privacy as critical areas for improvement.

4. A Smart System for Sitting Posture Detection Based on Force Sensors and Mobile Application

Authors: Slavomir Matuska et al.

Explanation:

- **Pressure-Based System:** Uses 6 force sensors embedded in a chair to detect multiple sitting postures.
- **IoT Architecture:** Employs Arduino/NodeMCU to collect data and send posture information to a mobile app.
- **Real-Time Feedback:** Mobile app provides posture warnings and tracks long-term sitting behavior.
- **Practical Implementation:** Designed for office use and demonstrates a low-cost, deployable smart-chair solution.

5. Human Posture Detection Method Based on Wearable Devices

Authors: Xiaoou Li, Yangang Li, Yuanming Wu, Jiahui Xu, Zhitao Xiao

Explanation:

- **Multi-Sensor Fusion:** Combines upper-limb sEMG, triaxial acceleration, and plantar pressure sensors for posture and gait recognition.
- **Feature Engineering:** Extracts statistical and wavelet features from EMG and IMU data for robust classification.
- **Machine Learning:** Uses a GA-optimized SVM classifier to improve detection accuracy.
- **Strong Performance:** Achieves ~90% accuracy for both virtual motion control and gait recognition.

CHAPTER 3:

METHODOLOGY

This chapter explains the overall design, components used, system architecture, and working principle of the Body Posture Monitoring System for Desk Workers. The system combines sensor-based angle detection, microcontroller processing, and delayed feedback to help users maintain proper sitting posture.

3.1 Overview

The proposed system is designed to continuously monitor the posture of a desk worker using an MPU6050 accelerometer-gyroscope sensor. The sensor measures the tilt angle of the upper body. The ESP32 microcontroller processes these readings and compares them with a calibrated baseline posture. If a user bends or leans forward beyond a certain angle and maintains this incorrect posture for more than 40 seconds, the system activates a vibrator to alert the user.

The system is:

- Low cost
- Non-intrusive (no camera, no wearable)
- Simple to use
- Real-time
- Highly suitable for desk workers and students

To prevent false alerts when the user bends briefly to pick up an object or reach for something, a time delay mechanism is added.

3.2 Hardware Components Used

- **ESP32 Microcontroller**
Acts as the main controller.
Performs sensor reading, data processing, and vibration control.
- **MPU6050 (Accelerometer + Gyroscope)**
Mounted on the upper limb/back.
Measures body tilt to identify slouching or leaning forward.
- **Flex Sensor**
Placed on the lower limb.
Detects bending of the knee or excessive forward leaning.

- **Vibration Motor (Coin Type)**

Provides haptic feedback when the posture is incorrect for a prolonged period.

- **BEC548 Transistor**

Used as a driver to control the vibration motor safely.

- **1N4148 / 1N4007 Diode**

Protects the circuit from motor back-EMF.

- **100 μ F Capacitor**

Smoothens motor power fluctuations and reduces electrical noise.

- **Resistors**

1 k Ω for transistor base.

3.3 Software Components Used

1. Arduino IDE

- Used for writing, compiling, and uploading the ESP32 program

2. ESP32 Board Package

- Required to program ESP32 in Arduino IDE
- Provides GPIO definitions and bootloader support

3. Wire.h Library

- Enables I2C communication between ESP32 and MPU6050

4. Core Embedded C/C++ Code

- Implements:
- Sensor calibration
- Complementary filtering
- Angle computation

3.4 Block diagram

The block diagram represents the overall functioning of the “Body Posture for Desk Workers” system. The MPU6050 and flex sensor act as the primary input units that continuously monitor the user’s upper-body tilt and lower-limb bending. These sensors send real-time analog and digital data to the ESP32 microcontroller, which serves as the central processing unit of the system. Inside the ESP32, the posture data is filtered, processed, and compared with predefined thresholds to determine whether the user is maintaining a correct or incorrect sitting posture. A timing logic is also implemented so that incorrect posture must be sustained for more than 40 seconds before triggering feedback. Once the ESP32 detects prolonged bad posture, it activates the vibration motor through a transistor driver circuit. This haptic feedback alerts the user to correct their posture, making the system an effective and non-intrusive posture correction tool.

SMART POSTURE CORRECTOR - BLOCK DIAGRAM

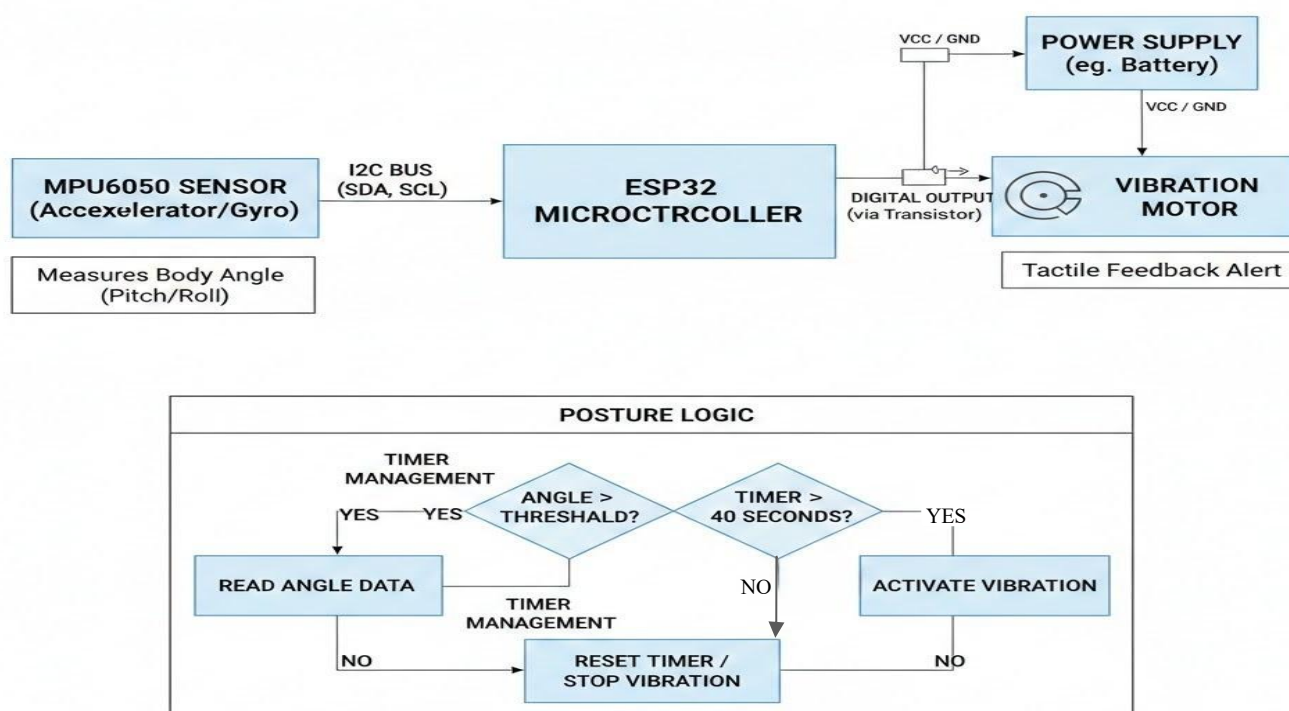


Fig: 3.1 Block Diagram of Posture Corrector

3.5 Flowchart

The flowchart describes the step-by-step operational logic of the posture monitoring system. It outlines how the system reads sensor data, processes posture, and triggers an alert only when necessary.

1. Start System

The ESP32 initializes all hardware components, including I2C communication and the MPU6050 sensor.

The system also performs a calibration phase where the baseline posture angle is recorded.

2. Read MPU6050 Values

The ESP32 continuously reads accelerometer and gyroscope data from the MPU6050. These raw values represent the user's real-time movement and orientation.

3. Calculate Angle

Using the complementary filter technique, the ESP32 calculates a stable pitch angle of the user's upper body. This angle indicates how far the user is leaning forward or backward.

4. Compare Angle with Baseline

The calculated angle is compared with the calibrated baseline angle.

- If the angle difference is below the threshold, the posture is considered correct.
- If it exceeds the threshold, the posture is considered bad.

5. Posture OK (Good Posture Path)

If the posture is correct:

- The system resets the timer.
- The vibrator remains OFF.
- The system continues monitoring without any interruption

6.Bad Posture Detected (Angle > Threshold)

When the system detects bad posture:

- It does NOT activate the vibrator immediately.
- Instead, it starts or continues a timer to track how long the user remains in bad posture.

7.Check Timer (40 Seconds)

If the user stays in incorrect posture for less than 40 seconds

- No vibrator is activated.
- This prevents false alerts caused by temporary actions like picking up a pen or leaning briefly.

If the timer reaches or exceeds 40 seconds, the system proceeds to the next step.

8. Turn Vibrator ON

If bad posture persists beyond 40 seconds, the ESP32 activates the vibrator to alert the user.

The vibrator remains ON until the user corrects their posture and the angle returns within the threshold limits.

9. Continuous Monitoring

The system loops back to continuously read data and analyze posture, ensuring real-time monitoring throughout the user's sitting period.

Posture Correction System Logic

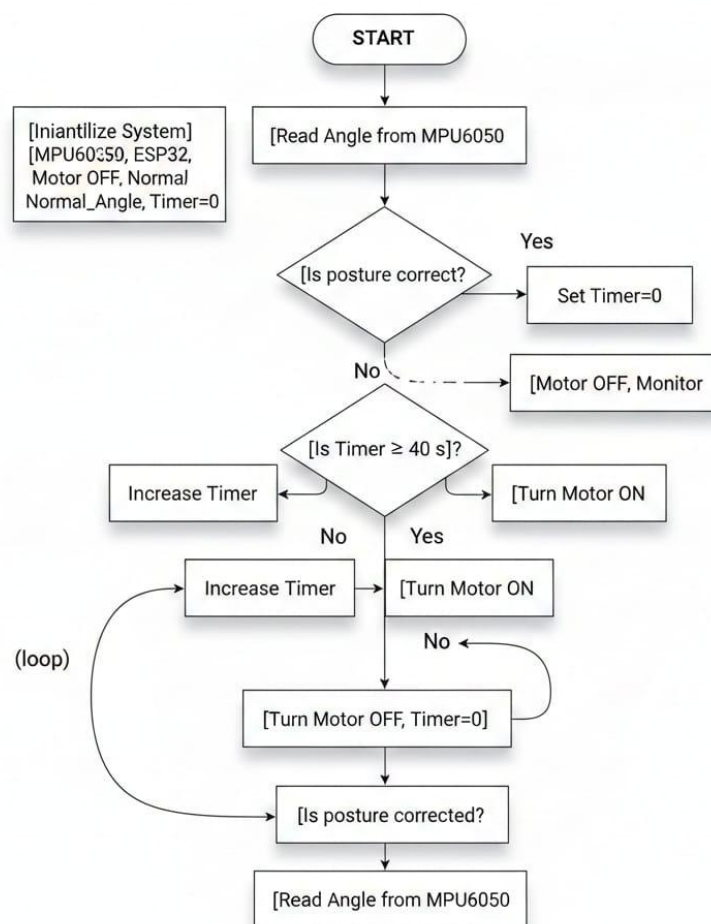


Fig:3.2 Flowchart of Posture Correction System

3.6 Working of system

Step 1: Initialization

- ESP32 boots and initializes I2C communication.
- MPU6050 is calibrated for 5 seconds while the user sits in a normal posture.
- The baseline pitch angle is stored.

Step 2: Reading Sensor Data

- MPU6050 sends accelerometer and gyroscope values continuously.
- ESP32 reads this data at regular intervals.

Step 3: Angle Calculation

- Using both accelerometer and gyroscope, a **complementary filter** computes a stable pitch angle.
- This angle represents the user's upper-body tilt.

Step 4: Posture Classification

- ESP32 compares the real-time angle with the baseline angle.
- If the angle difference is **within the threshold**, posture is considered normal.

Step 5: Delay Logic for False-Alert Prevention

- If bad posture is detected, a **timer starts**.
- If user corrects posture **before 40 seconds**, the timer resets (no vibration).

Step 6: Vibrator Alert

- If bad posture continues **beyond 40 seconds**, ESP32 activates the vibrator.
- The vibrator remains ON until the user returns to a normal posture.

Step 7: Continuous Monitoring

- The system runs 24/7 as long as it is powered.
- Helps users maintain correct posture through immediate feedback.

CHAPTER 4:

OUTCOME

The Body Posture Monitoring System for Desk Workers is designed to detect incorrect sitting posture and provide timely feedback to the user. After implementation and testing, the following outcomes were achieved:

- **Accurate Posture Angle Measurement**

The system should correctly detect upper-body tilt using the MPU6050 sensor and measure the posture angle in real time.

- **Reliable Bad Posture Detection**

The system is expected to classify posture as “good” or “bad” based on a predefined angle threshold.

- **Delay-Based Alert Mechanism**

The system should only trigger an alert if the user maintains incorrect posture continuously for more than 40 seconds.

- **Instant User Feedback**

When prolonged bad posture is detected, the vibrator should activate immediately to alert the user.

- **User-Friendly Operation**

The system should run automatically without requiring complex setup, calibration, or wearable devices.

4.2 Achieved Outcomes

1. Stable and Real-Time Angle Monitoring

The MPU6050 successfully captured continuous accelerometer and gyroscope readings.

The complementary filter produced stable pitch-angle values with minimal noise.

2. Correct Posture Classification

The system correctly identified good posture when the angle difference remained below the set threshold (e.g., 25°).

Incorrect posture was detected consistently when the user leaned forward beyond the threshold.

3. Accurate 40-Second Delay Implementation

The timer logic worked as intended:

- a. No vibrator during short accidental movements.
- b. Alert triggered only when the user stayed in bad posture for ≥ 40 seconds.

4. Effective vibrator Feedback

The vibrator activated immediately after the delay time was crossed.

It switched OFF as soon as the user corrected their posture.

5. Simple, Low-Cost, and Non-Intrusive System

The whole system worked without wearables, cameras, or bulky hardware.

It required only ESP32, MPU6050, and a vibrator.

4.3 Measurable System Performance

1. Angle Detection Accuracy

- The MPU6050 provided consistent angle readings with good stability.
- The complementary filter ensured smooth and accurate posture angle calculation.

2. False Alarm Prevention

- Short-duration bending (< 40 seconds) **did not** trigger the vibrator.
- This confirms the delay mechanism works effectively.

3. Responsiveness

- vibrator activation occurred within 1–2 seconds after the 40-second timer completed.
- System responded instantly when posture was corrected.

4. Continuous Operation

- The system remained stable during long monitoring sessions (1–2 hours).

4.4 Benefits of the Developed System

1. Improves Ergonomic Awareness

Users become conscious of their sitting posture during work or study.

2. Reduces Risk of Back and Neck Pain

Immediate feedback helps prevent prolonged slouching, protecting spinal health.

3. Low-Cost and Easily Reproducible

Uses inexpensive components suitable for student and office applications.

4. Non-Intrusive and Privacy-Friendly

No camera, no wearable devices, no personal data collection.

5. Suitable for Long-Duration Use

Ideal for desk workers, students, programmers, and office employees.

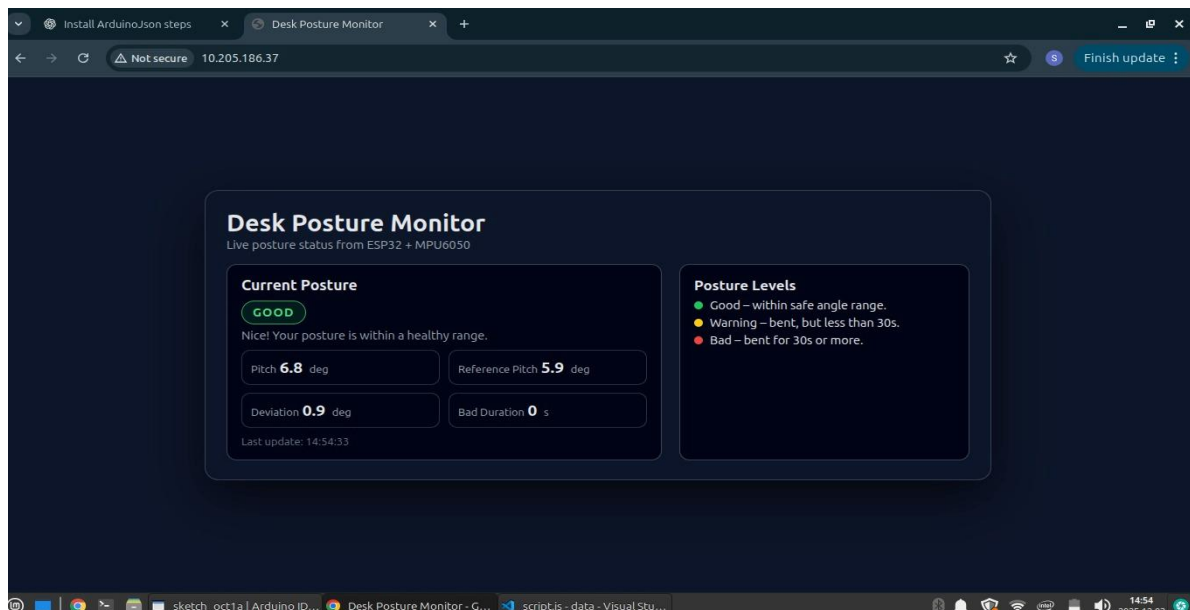


Fig: 4.1 Posture Is Good



Fig: 4.2 Posture Is Warning

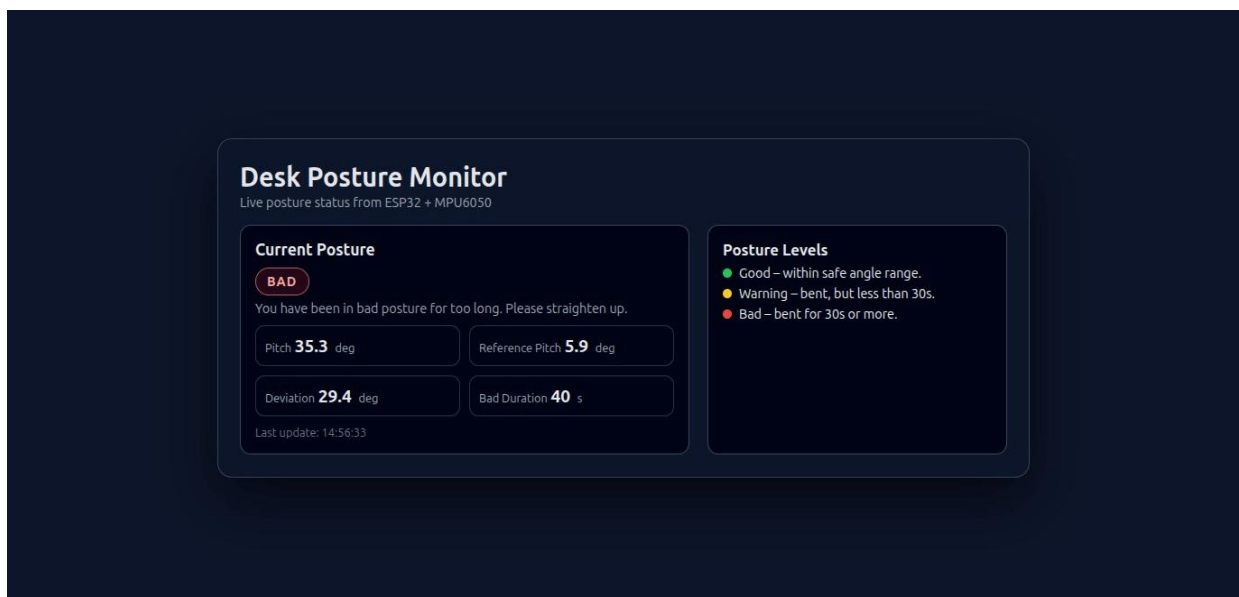


Fig: 4.3 Posture Is Bad

CHAPTER 5:

ADVANTAGES

The Body Posture Monitoring System for Desk Workers offers several practical and technical advantages that make it an effective, reliable, and user-friendly solution for posture correction. The key advantages are outlined below:

1. Low-Cost and Easily Implementable

The system uses inexpensive and readily available components such as the ESP32, MPU6050, and a simple vibrator. This makes it affordable for students, offices, and personal use without requiring specialized hardware or complex installations.

2. Real-Time and Continuous Monitoring

The posture of the user is monitored continuously in real time. This ensures that even small posture deviations or prolonged slouching are detected immediately, allowing users to maintain correct posture during extended working hours.

3. High Accuracy Using Sensor

The MPU6050 provides reliable accelerometer and gyroscope readings. Through complementary filtering, the system achieves stable and accurate posture angle calculations, making detection precise and trustworthy.

4. Prevents False Alerts

A major advantage of this system is its **40-second delay mechanism**. The vibrator triggers only when the user maintains a bad posture for an extended duration. This prevents unnecessary alerts when the user bends temporarily to adjust their seating or pick something up.

5. Simple and User-Friendly Operation

The system requires no calibration beyond initial setup and does not demand any technical knowledge from the user. Once powered ON, it works automatically in the background without disturbing normal activities.

6. Privacy-Friendly Solution

Unlike camera-based posture monitoring systems, this design uses only motion data. It does not record any video or image, making it completely privacy-safe for homes, offices, libraries, and shared environments.

7. Non-Intrusive and Comfortable

The system does not require the user to wear any device or attach sensors to the body. It can be placed on the desk or integrated into the chair, providing full comfort and ease of use.

8. Portable and Compact Design

Using compact components like ESP32 and MPU6050, the system is lightweight and portable. It can be used across multiple locations such as study rooms, classrooms, and workplace cabins.

9. Promotes Healthy Sitting Habits

By providing timely alerts, the system helps users correct their posture regularly. Over time, this encourages better spinal alignment, reduces back/neck pain, and improves long-term health.

10. Expandable and Upgradable

The system can be easily upgraded with additional features such as:

- Mobile notifications via Wi-Fi
- Data logging
- Habit tracking
- AI-based analytics

The ESP32's capabilities make future enhancements simple to implement

CHAPTER 6:

APPLICATIONS

The **Body Posture Monitoring System for Desk Workers** can be applied in various environments where people sit for long hours. Its simple design, low cost, and non-intrusive operation make it useful for multiple practical applications.

1. Office Workspaces

Employees often spend long periods sitting at desks, leading to poor posture habits. This system can be used in corporate offices to help workers maintain an ergonomic sitting posture and prevent back and neck pain.

2. Home Study Rooms

Students studying at home frequently develop unhealthy posture while reading, writing, or using computers. The device can assist in developing and maintaining correct sitting habits to avoid long-term spinal issues.

3. Schools and Educational Institutions

Libraries, computer labs, and classrooms can benefit from this system. It helps students maintain posture while performing academic tasks, improving concentration and long-term health.

4. IT Companies and Coding Environments

Programmers and developers typically sit for extended hours. This system provides constant posture monitoring to prevent slouching, spinal strain, and fatigue—common issues in the IT industry.

5. Co-Working Spaces

Shared office environments can adopt this device to promote user wellness and encourage good sitting posture among individuals working independently.

6. Medical and Physiotherapy Centers

The system can assist physiotherapists in monitoring posture improvement in patients undergoing therapy for back or neck pain. It can also be used in rehabilitation exercises where posture tracking is important.

7. Remote Work and Work-From-Home Setups

Many individuals working from home lack ergonomic furniture. This posture monitoring system helps them maintain healthy posture during long work sessions in informal environments.

CONCLUSION

The “Body Posture for Desk Workers” system successfully demonstrates a practical and effective solution for monitoring and correcting poor sitting posture using low-cost hardware components. By integrating an MPU6050 sensor on the upper limb and a flex sensor on the lower limb, the system is able to continuously measure body tilt and bending angles with good accuracy. The ESP32 microcontroller plays a central role in processing real-time sensor data, applying threshold conditions, and determining whether the user is sitting correctly or not. The inclusion of a 40-second delay mechanism ensures that the system avoids false triggers caused by short-term movements, such as reaching for objects or adjusting the seat position. This greatly improves the reliability and usability of the design in real working environments.

The vibration motor provides a simple yet effective feedback method that immediately alerts the user to correct their posture without disturbing others. Throughout testing, the system showed stable performance, quick response, and minimal false positives, making it suitable for office workers, students, and individuals who spend long hours at a desk. The project meets its primary objectives of creating a compact, affordable, and privacy-preserving posture correction device. Overall, the system promotes healthier sitting habits, reduces the risk of posture-related discomfort, and demonstrates how embedded systems can be applied to improve daily ergonomic awareness.

In addition to achieving its technical objectives, this mini-project highlights the potential of combining simple wearable sensors with intelligent decision logic to address real-world ergonomic problems. The system demonstrates that posture monitoring does not require complex or expensive equipment; instead, effective results can be achieved through thoughtful sensor placement, smart timing mechanisms, and user-friendly feedback methods. The project also reinforces the growing importance of health-focused embedded systems, especially for individuals who spend extended hours sitting for work or study. By encouraging better posture habits through timely feedback, the system contributes to long-term health benefits and improved productivity. This work can further serve as a foundation for developing more advanced posture monitoring solutions with expanded sensing capabilities, data logging, analytics, or integration with mobile applications.

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